Broiler farm documentation (Ciobanu Alin-Viorel)

1. Contents

2. Subsystem Use cases

A. *UC-01: Creating the Chicks Reception*

Use Case ID: UC-01
Use Case Name: Creating the Chicks Reception
Actor: Farm Manager
Preconditions:

- Delivery notice received from Hatchery subsystem
- Target poultry house is EMPTY or available
- Farm Manager is authenticated

Main Flow:

1. System receives delivery notice from Hatchery subsystem
2. System auto-creates draft ChickReception record
3. Farm Manager reviews incoming delivery details
4. Farm Manager selects target Poultry House for each line of chicks lots
5. System validates house capacity and availability
6. Farm Manager inputs reception details:
    - Reception date and time
    - Actual quantity received
    - Number of chicks alive
    - Number of DOA (Dead on Arrival)
    - Number of weak chicks
    - Quality assessment
    - Transport conditions
7. Farm Manager confirms reception
8. System creates new ChicksLot record
9. System updates Poultry House status to OCCUPIED
10. System sends confirmation to Hatchery subsystem
11. System displays success message

Alternative Flows:

- **A1: Insufficient Capacity**
  - 5a. If house capacity < received quantity
  - System displays error message
  - Farm Manager must select different house
- **A2: House Not Available**
  - 5a. If house status is not EMPTY
  - System suggests alternative houses
  - Farm Manager selects different house

Postconditions:

- ChickReception record created
- New ChicksLot initialized
- Poultry House status updated to OCCUPIED
- Hatchery receives confirmation

Business Rules:

- One active lot per poultry house
- Reception line quantity cannot exceed house capacity
- Lot number auto-generated: FARM-HOUSE-YYYY-WW

B. UC-02: Editing the Chicks Reception

Use Case ID: UC-02
Use Case Name: Editing the Chicks Reception
Actor: Farm Manager
Preconditions:

- ChickReception record exists
- Reception is not finalized (within 24 hours)
- Farm Manager is authenticated

Main Flow:

1. Farm Manager searches for reception record
2. System displays list of recent receptions
3. Farm Manager selects reception to edit

4. System displays reception details
5. Farm Manager modifies editable fields:
   - Reception date/time (within same day)
   - Quality assessment details
   - Transport conditions
   - Notes
6. Farm Manager saves changes
7. System validates modified data
8. System recalculates affected metrics
9. System updates ChicksLot if quantities changed
10. System logs modification with timestamp and user
11. System displays success message

Alternative Flows:

- **A1: Reception Already Finalized**
  - 4a. If reception older than 24 hours
  - System displays "Requires manager approval" message
  - Farm Manager must request approval workflow
- **A2: Invalid Quantity Change**
  - 7a. If new quantity conflicts with existing mortality records
  - System displays validation error
  - Farm Manager must adjust related documents first

Postconditions:

- ChickReception record updated
- Related ChicksLot updated if quantities changed
- Audit trail recorded

Business Rules:

- Cannot modify reception after 24 hours without approval
- Cannot change house assignment after reception

C. *UC-03: Monitoring the Chick Stock for All Farm*

Use Case ID: UC-03
Use Case Name: Monitoring the Chick Stock for All Farm
Actor: Farm Manager
Preconditions:

- Farm Manager is authenticated
- At least one active ChicksLot exists

Main Flow:

1. Farm Manager accesses monitoring dashboard
2. System retrieves all active lots across all poultry houses
3. System displays farm-wide overview:
   - Total birds alive
   - Total capacity utilization
   - Active lots count
   - Houses occupied/empty
4. Farm Manager can filter by:
   - Poultry house
   - Date range
   - Lot status
   - Performance metrics
5. System displays detailed metrics for each lot:
   - Current quantity
   - Days in farm
   - Cumulative mortality rate
   - Latest average weight
   - Feed Conversion Ratio (FCR)
   - Health status
   - Expected delivery date
6. Farm Manager can drill down into specific lot
7. System displays lot history:

    - Reception details
    - Daily consumption trends
    - Mortality trends
    - Treatment history
    - Observation sheets
8. Farm Manager can export reports
9. System generates PDF/Excel report

Alternative Flows:

- **A1: No Active Lots**
    - 2a. If no active lots found
    - System displays "No active lots" message
    - Shows historical data option
- **A2: Performance Alert**
    - 5a. If any lot exceeds threshold (mortality, FCR)
    - System highlights lot in red
    - Displays alert icon with details

Postconditions:

- Dashboard viewed
- Reports generated if requested

Business Rules:

- Alerts triggered if mortality > expected + 2%
- FCR calculated from latest observation sheet

D. UC-04: Create the Consumption Sheet

Use Case ID: UC-04
Use Case Name: Create the Consumption Sheet
Actor: Logistics Manager (Primary)
Preconditions:

- Active ChicksLot exists in poultry house
- Logistics Manager is authenticated

- Consumables available in stock
- No consumption sheet exists for current date

Main Flow:

1. System auto-creates daily ConsumptionSheet at midnight for each active lot
2. Logistics Manager monitors stock levels daily
3. Logistics Manager checks if consumption sheet exists for today
4. If no sheet exists:
    - Logistics Manager accesses consumption sheet creation
    - System displays pre-filled template with:
        - Date
        - Poultry house
        - Current birds alive count
5. Logistics Manager records consumption for each feeding:
    - Consumable type (feed type, additives)
    - Quantity consumed
    - Feeding time
    - Notes (if any)
6. System calculates consumption per bird automatically
7. Logistics Manager can add multiple consumption lines throughout day
8. Logistics Manager finalizes sheet at end of day
9. System validates total consumption against available stock
10. System auto-decrements ConsumableStock for each line
11. System calculates daily totals
12. System updates lot consumption history
13. System displays success message

Alternative Flows:

- A1: Sheet Already Exists
    - 4a. If consumption sheet already exists for today
    - System displays existing sheet

    - [Include UC-08: Update the Consumption Sheet]
    - Continue with modifications
- A2: Insufficient Stock
    - 9a. If consumption exceeds available stock
    - System displays error with current stock levels
    - Logistics Manager must adjust quantities or receive new stock first
    - [Include UC-09: Create Reception of Consumables]
- A3: Delete Erroneous Entry
    - At any step, Logistics Manager can delete consumption line
    - System reverses stock deduction
    - Requires confirmation

Postconditions:

- ConsumptionSheet created
- ConsumableStock updated
- Daily metrics calculated
- Data available for observation sheet

Business Rules:

- One consumption sheet per day per house
- Can edit/delete within 24 hours freely
- After 24 hours requires manager approval
- Cannot consume expired consumables
- Minimum consumption per bird thresholds enforced

Includes:

- UC-08: Update the Consumption Sheet (if sheet already exists)
- UC-09: Create Reception of Consumables (if stock insufficient)

E. UC-05: Create the Observations Sheet

Use Case ID: UC-05
Use Case Name: Create the Observations Sheet

Actor: Farm Manager (Primary), Worker (Secondary)
Preconditions:

- Active ChicksLot exists
- At least 7 days since reception or last observation
- Farm Manager is authenticated

Main Flow:

1. **[Time Event: Weekly - Every 7 days]** System auto-creates weekly ObservationSheet
2. Farm Manager accesses observation sheet for their house
3. System auto-populates calculated metrics:
   - Starting bird count (from last week or reception)
   - Ending bird count (current - weekly mortality)
   - Total mortality count and percentage
   - Average daily mortality
   - **[Include UC-06: Send the Consumptions Informations]**
   - Total feed consumed (from ConsumptionSheets)
   - Total water consumed
4. Farm Manager performs manual measurements:
   - Random weight sampling (minimum 10 birds)
   - Records individual weights
   - System calculates average weight
5. Farm Manager inputs observational data:
   - Health observations (appearance, activity)
   - Behavioral notes (feeding, drinking patterns)
   - Environmental conditions (temperature, humidity)
   - Any concerns or issues
6. System calculates performance metrics:
   - Weight gain since last observation
   - Feed Conversion Ratio (FCR)
   - Average Daily Gain (ADG)

    - Mortality rate comparison to expected
7. Farm Manager saves observation sheet
8. System validates all required fields completed
9. Farm Manager submits for approval
10. **Farm Manager (self) approves the observation sheet**
11. System marks sheet as APPROVED
12. System updates lot statistics
13. **System evaluates if growing process is complete:**
    - **If lot age >= 35 days AND no withdrawal periods active:**
        - System flags lot as eligible for delivery
        - **Farm Manager can proceed to [UC-10: Create Delivery Notice]**
    - **If not ready:**
        - **System returns to daily monitoring loop**
        - Continue daily activities (consumption, mortality, treatment tracking)
14. System displays success message

Alternative Flows:

- **A1: Incomplete Data**
    - 8a. If required fields missing
    - System highlights missing fields
    - Farm Manager must complete before submission
- **A2: Performance Out of Range**
    - 6a. If metrics significantly deviate from expected
    - System displays warning
    - Farm Manager must add explanatory notes
    - Flags for investigation
- **A3: Missing Consumption Data**
    - 3a. If ConsumptionSheets incomplete for the week
    - **[Include UC-06: Send Consumption Informations with warnings]**
    - System displays warning about missing data

    - Farm Manager can proceed with manual entry

Postconditions:

- ObservationSheet created and approved
- Lot performance metrics updated
- Data available for delivery planning
- Historical record stored
- **Decision point reached: Continue growing OR prepare for delivery**

Business Rules:

- Created weekly on configured day (e.g., every Sunday)
- Minimum 10 birds for weight sampling
- Weight variance >20% triggers alert
- FCR > 2.0 triggers investigation
- Manager approval required before finalization
- **Observation sheet triggers evaluation of delivery eligibility**

Includes:

- UC-06: Send the Consumptions Informations (automatic data aggregation)

Leads To:

- **Decision Point:** Is final schedule of growing process?
    - **YES** → UC-10: Create the Delivery Notice
    - **NO** → Return to daily activities loop

F. *UC-06: Send the Consumptions Informations*

Use Case ID: UC-06
Use Case Name: Send the Consumptions Informations
Actor: System (Automated)
Preconditions:

- ConsumptionSheets exist for the observation period
- ObservationSheet is being created

Main Flow:

1. System retrieves all ConsumptionSheets for the lot within date range

2. System aggregates consumption data:
    - Total feed consumed by type (starter, grower, finisher)
    - Total additives/supplements
    - Total water consumed
3. System calculates:
    - Total consumption in kg
    - Average daily consumption
    - Consumption per bird
4. System transfers aggregated data to ObservationSheet
5. System links source ConsumptionSheets for traceability
6. System marks data as "auto-calculated"

Alternative Flows:

- **A1: Missing Consumption Data**
    - 1a. If any days missing consumption sheets
    - System flags missing days
    - Displays warning in observation sheet
    - Allows manual entry with notation

Postconditions:

- Consumption data integrated into observation
- Source data linked
- Ready for FCR calculation

Business Rules:

- Includes all consumption types
- Excludes deleted/rejected sheets
- Only includes approved consumption sheets

G. UC-07: Creating the Treatment Sheet

Use Case ID: UC-07
Use Case Name: Creating the Treatment Sheet
Actor: Farm Veterinarian (Primary), Worker (Secondary)

Preconditions:

- Active ChicksLot exists
- **Health issue identified** (conditional - not daily routine)
- Veterinarian is authenticated
- Medications available in stock

Main Flow:

1. [Conditional Daily Activity - Only if health issue identified]
2. Decision: Has health issue been identified?
   - **NO** → Skip treatment sheet creation
   - **YES** → Continue
3. Veterinarian diagnoses health issue
4. Veterinarian accesses treatment sheet creation
5. Veterinarian adds treatment lines:
   - Medication/vaccine from consumables
   - Dosage and unit
   - Administration method (water, feed, injection)
   - Duration (number of days)
   - Start date
   - Withdrawal period (days before slaughter allowed)
6. System validates:
   - Medication availability in stock
   - Dosage within safe ranges
   - No conflicting treatments
7. System calculates:
   - End date (start + duration)
   - Slaughter allowed date (end + withdrawal)
   - Total quantity needed
8. System checks stock sufficiency
9. Veterinarian saves treatment sheet
10. System creates draft treatment sheet

11. Veterinarian reviews and approves
12. System activates treatment
13. System blocks lot from delivery until withdrawal period ends
14. System creates scheduled ConsumptionSheet entries for treatment duration
15. System sends notification to workers
16. System displays success message

Alternative Flows:

- **A1: Insufficient Medication Stock**
  - 8a. If medication quantity insufficient
  - System displays current stock levels
  - Veterinarian must adjust dosage or wait for stock
- **A2: Conflicting Treatment**
  - 6a. If another active treatment exists with drug interactions
  - System displays warning
  - Veterinarian must resolve conflict
- **A3: Worker Creates Draft**
  - 2a. Worker can create draft treatment for vet review
  - Worker enters observations only
  - Vet must review and add prescriptions

Postconditions:

- TreatmentSheet created and activated
- Lot delivery blocked during treatment + withdrawal
- Consumption entries scheduled
- Workers notified
- Stock reserved for treatment
- If treatment created: Extends UC-08 (Update Consumption Sheet) automatically

Business Rules:

- Only veterinarian can approve treatments
- One treatment sheet per day per house

- Withdrawal period must be respected
- Cannot delete active treatment (can only deactivate)
- Medication batch numbers must be recorded

Extends:

- Create the Consumption Sheet (auto-creates medication consumption entries)
- Update the Consumption Sheet (updates daily with treatment dosages)

H. UC-08: Update the Consumption Sheet

Use Case ID: UC-08
Use Case Name: Update the Consumption Sheet
Actor: Logistics Manager (Primary), System (Automated from Treatment)
Preconditions:

- ConsumptionSheet exists (created today)
- Logistics Manager is authenticated OR automated process triggered

Main Flow:

1. **[Triggered by UC-04 Alternative Flow A1 OR UC-07 Extension]**
2. Logistics Manager or system accesses existing consumption sheet
3. System displays current consumption lines
4. **If manual update (Logistics Manager):**
    - Logistics Manager edits existing line quantities
    - Logistics Manager adds new consumption lines
    - Logistics Manager corrects errors
5. **If automated update (from Treatment - UC-07):**
    - System adds medication consumption line
    - System calculates quantity based on treatment dosage
    - System marks as "auto-generated from treatment"
6. System validates changes:
    - Stock availability
    - Quantity reasonableness
    - No expired consumables
7. System recalculates:
    - Total consumption
    - Consumption per bird
    - Updated stock levels
8. System updates ConsumableStock
9. System logs modification with timestamp and source
10. System displays success message

Alternative Flows:

- **A1: Edit After 24 Hours**
  - 1a. If sheet older than 24 hours
  - System requires Farm Manager approval
  - Creates approval request
  - Manager must approve before changes applied
- **A2: Stock Reversal Needed**
  - 4a. If editing/deleting existing line
  - System reverses original stock deduction
  - Then applies new stock deduction
- **A3: Validation Error**
  - 6a. If new quantities invalid
  - System displays specific error
  - Logistics Manager must correct before saving
- **A4: Insufficient Stock for Update**
  - 6a. If updated quantities exceed available stock
  - System displays error
  - **[Include UC-09: Create Reception of Consumables]**
  - Logistics Manager must receive stock first

Postconditions:

- ConsumptionSheet updated
- Stock levels adjusted
- Audit trail recorded
- Related calculations updated

Business Rules:

- Within 24h: free edit/delete
- After 24h: requires manager approval
- Cannot modify sheet from finalized observation period
- System updates take precedence over manual in case of conflict

Extended By:

- UC-07: Creating the Treatment Sheet (auto-updates with medication consumption)

I. *UC-09: Create the Reception of Consumables*

Use Case ID: UC-09
Use Case Name: Create the Reception of Consumables
Actor: Logistics Manager (Primary), Worker (Secondary)
Preconditions:

- **Consumable stock is below optimal level OR supplier delivery expected**
- Logistics Manager is authenticated

Main Flow:

1. **[Triggered by daily stock monitoring OR scheduled supplier delivery]**
2. **Logistics Manager checks stock levels**
3. **System displays current stock status:**
    - Items below reorder point highlighted
    - Expected consumptions for next 7 days
    - Alerts for low/critical stock
4. **Decision: Is stock level optimal?**
    - **NO** → Continue with reception process
    - **YES** → Exit use case
5. Logistics Manager creates new ConsumableReception
6. System displays reception form
7. Logistics Manager enters header information:
    - Supplier name
    - Reception date
    - Purchase order reference (if exists)
    - Receiving warehouse/location
8. Logistics Manager adds reception lines for each item:
    - Selects consumable from catalog
    - Enters quantity received
    - Records unit price
    - Enters batch/lot number
    - Records manufacturing date
    - Records expiration date
    - Specifies storage location
9. Worker performs quality inspection
10. Worker marks quality status for each line:
    - ACCEPTED: good quality
    - REJECTED: damaged, wrong item
    - QUARANTINE: needs further inspection
11. System validates:
    - Quantities match PO (within tolerance) if PO exists
    - Expiration dates in future
    - Prices reasonable
12. System calculates total amount
13. Logistics Manager reviews reception
14. Logistics Manager approves reception
15. **For ACCEPTED items:**
    - System creates/updates ConsumableStock records
    - System adds quantity to stock
    - System records batch and expiration info
16. **For REJECTED items:**
    - System creates rejection record
    - System generates supplier return notice
17. System updates purchase order status (if external PO exists)
18. System generates reception report
19. **System returns to consumption sheet management**
20. System displays success message

Alternative Flows:

- **A1: Stock is Optimal**
    - 4a. If all stock levels are adequate
    - Logistics Manager skips reception
    - **Proceeds directly to [UC-04: Create Consumption Sheet]**
- **A2: Quantity Discrepancy**
    - 11a. If received quantity differs from PO significantly
    - System displays warning
    - Logistics Manager must add explanation notes
    - May require supplier contact
- **A3: Quality Issues**
    - 10a. If multiple items marked REJECTED
    - System creates incident report
    - Notifies quality assurance
    - Generates claim documents
- **A4: Emergency Reception (No PO)**
    - 5a. If no PO exists (emergency purchase)
    - System allows manual entry
    - Requires Farm Manager approval
    - Flags for accounting review
- **A5: Partial Reception**
    - 8a. If not all items delivered
    - Logistics Manager records only received items
    - System marks PO as "partially received"
    - Backorder created for missing items

Postconditions:

- ConsumableReception created and approved
- ConsumableStock updated for accepted items
- Supplier notified of rejections if applicable
- Purchase order updated (if exists)
- Inventory records current
- **Stock available for consumption sheet creation**

Business Rules:

- **Daily stock monitoring is mandatory for Logistics Manager**
- Reception must reference PO or be marked as emergency
- Cannot modify reception after approval
- Quality inspection required for all items
- FIFO (First In, First Out) enforced by expiration date
- Expired items automatically rejected
- Stock updates only for ACCEPTED items

Leads To:

- Returns to UC-04 (Create Consumption Sheet) or UC-08 (Update Consumption Sheet)

J. *UC-10: Create the Delivery Notice*

Use Case ID: UC-10
Use Case Name: Create the Delivery Notice
Actor: Farm Manager
Preconditions:

- Active ChicksLot exists
- **Latest ObservationSheet approved (from UC-05)**
- Lot meets minimum age requirement (typically 35-42 days)
- No active treatment withdrawal periods
- Farm Manager is authenticated

Main Flow:

1. **[Triggered by UC-05 decision: "Is final schedule of growing process?" → YES]**
2. **Farm Manager decides how to complete delivery notice:**
   - **Option A: Use data from last ObservationSheet (automatic)**
   - **Option B: Complete manually with raw data**
3. Farm Manager accesses delivery planning module
4. System displays eligible lots:
   - Age requirement met
   - No withdrawal period blocks
   - Status = GROWING or READY_FOR_DELIVERY
5. Farm Manager selects lot for delivery
6. System validates delivery eligibility:
   - Checks age >= minimum days (e.g., 35)
   - Verifies no active withdrawal periods
   - Confirms sufficient quantity available
7. **If Option A selected (automatic from observation):**
   - System retrieves data from latest ObservationSheet:
     - Average weight
     - Current bird count
     - Health status
     - FCR performance
8. **If Option B selected (manual raw data):**
   - Farm Manager manually enters:
     - Estimated quantity
     - Average weight (from manual weighing)
     - Health observations
   - System flags as "Manual Entry - Verify Data"
9. System pre-fills DeliveryNotice form:
   - Proposed delivery date (suggests next available slot)
   - Estimated quantity
   - Average weight
   - Total estimated weight
   - Quality grade based on metrics
10. Farm Manager can adjust:
    - Delivery date
    - Quantity (partial delivery allowed)

- Destination slaughterhouse
- Special instructions

11. Farm Manager adds logistics details:
    - Transport manager assignment
    - Preferred vehicle
    - Loading time
    - Handling requirements
12. System calculates:
    - Total estimated weight
    - Expected processing time
    - Vehicle capacity check
13. Farm Manager saves draft
14. System validates all business rules
15. Farm Manager submits for approval
16. System marks as PENDING_APPROVAL
17. Farm Manager (or senior manager) approves
18. System updates lot status to READY_FOR_DELIVERY
19. System notifies Transport Manager
20. **[Proceed to UC-11: Send Delivery Notice to Next Subsystem]**
21. System displays confirmation with delivery details

Alternative Flows:

- **A1: No Recent ObservationSheet Available**
    - 7a. If latest observation > 7 days old
    - System warns Farm Manager
    - Farm Manager must either:
        - Create new observation sheet first [Return to UC-05]
        - OR proceed with manual data entry (Option B)
- **A2: Withdrawal Period Active**
    - 6a. If any treatment has active withdrawal period
    - System displays blocking message with:
        - Treatment details
        - Withdrawal end date
        - Earliest allowed delivery date
    - Farm Manager cannot proceed
    - **Return to daily activities loop**
- **A3: Below Minimum Age**
    - 6a. If lot age < minimum required days
    - System displays warning
    - Shows current age and days remaining
    - Farm Manager can request exception approval
- **A4: Poor Performance Metrics**
    - 7a/8a. If FCR, mortality, or weight below thresholds
    - System displays warning alert
    - Suggests alternative delivery date
    - Farm Manager must add justification notes
    - Requires additional approval level
- **A5: Partial Delivery**
    - 10a. If Farm Manager reduces quantity

    - System splits lot into:
        - Delivery portion
        - Remaining portion
    - Creates new lot for remaining birds
    - Both tracked separately
- **A6: Approval Rejected**
    - 17a. If approver rejects delivery
    - System returns to draft with comments
    - Farm Manager must address issues and resubmit

Postconditions:

- DeliveryNotice created and approved
- Transport record created
- Lot status updated to READY_FOR_DELIVERY
- Transport Manager assigned and notified
- Delivery scheduled in calendar
- **Ready for transmission to Slaughterhouse**

Business Rules:

- Cannot deliver during withdrawal period (hard block)
- Minimum age: 35 days (configurable, can request exception)
- Maximum age: 50 days (quality deteriorates)
- Partial deliveries allowed (minimum 500 birds)
- **Prefer observation sheet data; manual entry requires justification**
- Requires farm manager approval
- One delivery notice can cover multiple houses (same lot batch)
- Average weight must be ≥ 1.8 kg (configurable)

Triggered By:

- UC-05: Create Observations Sheet (decision: growing process complete)

Leads To:

- UC-11: Send the Delivery Notice to Next Subsystem

K. UC-11: Send the Delivery Notice to Next Subsystem

Use Case ID: UC-11
Use Case Name: Send the Delivery Notice to Next Subsystem
Actor: Farm Manager (initiator), System (executor)
Preconditions:

- DeliveryNotice is approved

- Transport record created

- Scheduled delivery date/time set
- System integration with Slaughterhouse active

Main Flow:

1. Farm Manager confirms delivery notice is ready to send
2. System validates delivery notice completeness:
    - All required fields populated
    - Transport details confirmed
    - Approval status = APPROVED
3. System prepares data package for Slaughterhouse:
    - Delivery notice ID
    - Scheduled arrival date/time
    - Lot information (ID, age, breed)
    - Estimated quantity and weight
    - Average bird weight
    - Quality grade
    - Health status
    - Treatment history summary
    - Withdrawal period confirmations
    - Special handling instructions
    - Transport details
    - Contact information
4. System establishes connection to Slaughterhouse subsystem
5. System transmits delivery notice via API/integration layer
6. Slaughterhouse system receives and acknowledges
7. System updates DeliveryNotice status to SENT
8. System records transmission timestamp
9. System receives preliminary scheduling response from Slaughterhouse:
    - Assigned receiving dock
    - Confirmed time slot

    - Required documentation
10. System updates Transport record with Slaughterhouse details
11. System notifies Transport Manager with updated info
12. System creates delivery tracking record
13. System displays confirmation message with tracking details

Alternative Flows:

- **A1: Transmission Failure**
    - 5a. If connection to Slaughterhouse fails
    - System queues delivery notice for retry
    - System notifies Farm Manager of delay
    - System attempts retransmission (max 3 attempts)
    - If all fail, generates alert for manual intervention
- **A2: Slaughterhouse Rejects Delivery**
    - 6a. If Slaughterhouse system rejects (capacity issues)
    - System receives rejection message with reason
    - System notifies Farm Manager
    - System suggests alternative dates
    - Farm Manager must reschedule
- **A3: Schedule Conflict**
    - 9a. If requested slot unavailable
    - Slaughterhouse proposes alternative times
    - System presents options to Farm Manager
    - Farm Manager selects new time
    - System retransmits updated notice
- **A4: Missing Critical Data**
    - 2a. If validation finds missing information
    - System lists missing fields
    - Farm Manager must complete before sending
    - Returns to previous use case

Postconditions:

- DeliveryNotice transmitted successfully
- Slaughterhouse has scheduled receipt
- Transport details confirmed
- Status updated to SENT or IN_TRANSIT
- Tracking initiated
- All parties notified

Business Rules:

- Must send at least 24 hours before scheduled delivery
- Cannot modify after sending (must cancel and resend)
- Slaughterhouse must acknowledge within 2 hours
- If rejected, automatic reschedule workflow triggered
- All treatment/medication history must be included
- Withdrawal period compliance must be certified

Integration Points:

- **Outbound to Slaughterhouse:**
  - Delivery schedule
  - Lot details
  - Quality metrics
  - Health history
  - Treatment records
- **Inbound from Slaughterhouse:**
  - Acknowledgment
  - Dock assignment
  - Required prep instructions
  - Quality specifications
- **Real-time Updates:**
  - Capacity availability
  - Schedule changes

    - Special requirements

L. UC-012: Create the Mortality Sheet

Use Case ID: UC-012
Use Case Name: Create Daily Mortality Sheet
Actor: Worker (Primary), Farm Manager (Approver)

Preconditions:

- Active ChicksLot exists in the PoultryHouse
- Worker is authenticated and assigned to the poultry house
- Previous day's mortality sheet exists (except for first day after reception)

Main Flow:

1. **[Daily Activity - Parallel with Consumption and Treatment]**
2. Worker initiates daily mortality sheet creation
    - Worker accesses their assigned PoultryHouse
    - Worker selects "Create Mortality Sheet" for current date
    - System verifies no sheet already exists for current date
3. System auto-populates header information:
    - Date (current date)
    - PoultryHouse reference
    - Active ChicksLot reference
    - Reporting Employee (logged-in worker)
    - Current age of birds in days (calculated from reception date)
4. Worker records mortality events throughout the day:
    - For each mortality discovery, worker adds MortalitySheetLine:
        - Time discovered (timestamp)
        - Death count (number of birds found)
        - Cause of death (observation-based: respiratory, trauma, unknown, etc.)
        - Estimated weight of deceased birds (optional, for analysis)
        - Location in house (zone/section identifier)
        - Disposal method (BURIAL, INCINERATION, COMPOSTING)
    - Worker can add multiple lines throughout the day
5. System calculates mortality metrics:
    - Total daily mortality (sum of all line deathCounts)
    - Cumulative mortality (previous cumulative + today's total)
    - Current bird count (starting count - cumulative mortality)
    - Daily mortality rate (today's mortality / current population) × 100
    - Cumulative mortality rate (cumulative mortality / starting population) × 100
6. Worker finalizes and submits sheet:
    - Worker reviews all recorded mortality events
    - Worker adds any general observations or notes
    - System validates all required fields completed
    - Worker submits for manager approval
7. Farm Manager reviews and approves:

    - Manager accesses pending mortality sheets
    - Reviews mortality patterns and causes
    - Checks if rates are within acceptable range
    - Manager approves or rejects with comments
8. System finalizes approved sheet:
    - System marks sheet status as APPROVED
    - System updates ChicksLot statistics:
        - Current bird count
        - Cumulative mortality
        - Mortality rate trends
    - **Data available for weekly ObservationSheet (UC-05)**
    - System displays success confirmation

Alternative Flows: [All alternative flows remain the same as original]

Postconditions:

- MortalitySheet record created and approved
- MortalitySheetLines recorded with all mortality events
- ChicksLot statistics updated (current count, cumulative mortality)
- Historical mortality data preserved for analysis
- Mortality trends available for reporting
- **Data integrated into lot performance metrics**
- **Data available for UC-05 (Observation Sheet) weekly aggregation**

Business Rules: [All business rules remain the same]

Includes:

- Update Lot Statistics (automatically triggered on approval)
- Generate Mortality Alerts (triggered if thresholds exceeded)

Data Flows To:

- UC-05: Create Observations Sheet (mortality data aggregated weekly)

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3. Entities and the logic of them

Entity Definitions, relationships and business limitations

a) BaseEntity (Abstract)

- id: Long (PK)

- createdAt: DateTime

- createdBy: Employee

- updatedAt: DateTime

- updatedBy: Employee

- isActive: Boolean

- version: Integer (for optimistic locking)

b) Employee

- firstName: String

- lastName: String

- email: String (unique)

- phone: String

- role: Enum (FARM_MANAGER, LOGISTICS_MANAGER, TRANSPORT_MANAGER, WORKER, VETERINARIAN)

- hireDate: Date

- isActive: Boolean

- farm: BroilerFarm (FK)

c) BroilerFarm

- farmName: String

- location: String

- address: String

- capacity: Integer (total birds)

- licenseNumber: String

- manager: Employee (FK)

d) PoultryHouse

- houseNumber: String

- farm: BroilerFarm (FK)

- capacity: Integer (max birds)

- currentLot: ChicksLot (FK)

- area: Decimal (square meters)

- type: Enum (OPEN, CLOSED, SEMI_CLOSED)

- equipmentType: String

- status: Enum (ACTIVE, MAINTENANCE, EMPTY, OCCUPIED)

- currentOccupancy: Integer

Business rules:

- Cannot exceed capacity
- House number unique per farm
- Must be empty before new reception

e) ChicksLot

- lotNumber: String (unique)
- hatcherySource: String
- poultryHouse: PoultryHouse (FK)
- breed: String
- receptionDate: Date
- initialQuantity: Integer
- currentQuantity: Integer (calculated)
- expectedSlaughterDate: Date
- actualSlaughterDate: Date
- status: Enum (GROWING, READY_FOR_DELIVERY, DELIVERED, CLOSED)
- expectedMortalityRate: Decimal
- mortalitySheets: List<MortalitySheet>
- observationSheets: List<ObservationSheet>
- consumptionSheets: List<ConsumptionSheet>
- treatmentSheets: List<TreatmentSheet>

Calculated Fields:

- currentQuantity = initialQuantity - total mortality - total delivered
- actualMortalityRate = (total deaths / initialQuantity) * 100
- daysInFarm = currentDate - receptionDate
- averageWeight (from latest observation)

Business Rules:

- One active lot per poultry house at a time
- Lot number format: FARM-HOUSE-YYYY-WW (year-week)

f) ChicksReception

- hatcheryDeliveryNoticeId: String (from Hatchery system)

- receptionDate: DateTime

- farm: BroilerFarm (FK)

- receivingEmployee: Employee (FK)

- transportConditions: String

- documentReference: String

- receptionLines: List<ChicksReceptionLine>

- truckInfo: String

- totalQuantityReceived: Integer

- totalChicksAlive: Integer

- totalChicksDOA: Integer

- totalChicksWeak: Integer

Relationships:

- Has many ChicksReceptionLines

g) ChicksReceptionLine

- reception: ChickReception (FK)

- poultryHouse: PoultryHouse (FK)

- lot: ChicksLot

- quantity: Integer

- chicksAlive: Integer

- chicksDOA: Integer (Dead on Arrival)

- chicksWeak: Integer

- qualityGrade: Enum (A, B, C)

- notes: String

Business Rules:

- quantity = chicksAlive + chicksDOA + chicksWeak
- Creates initial ChicksLot from Delivery Notice from Hatchery , and after the creation of the reception , ChicksLot will have set the associated Poultry house
- Updates PoultryHouse occupancy

h) Consumable

- name: String

- type: Enum (FEED_STARTER, FEED_GROWER, FEED_FINISHER, MEDICATION, VACCINE, SUPPLEMENT, BEDDING, DISINFECTANT)

- category: String

- unitOfMeasure: Enum (KG, L, UNITS, BAGS)

- reorderPoint: Decimal

- standardPrice: Decimal

- supplier: String

- storageRequirements: String

- shelfLife: Integer (days)

i) ConsumableStock

- consumable: Consumable (FK)

- warehouse: Warehouse

- batchNumber: Integer

- quantityOnHand: Decimal

- reservedQuantity: Decimal

- availableQuantity: Decimal (calculated)

- lastRestockDate: Date

- manufacturingDate: Date

- expirationDate: Date

- status: ENUM(AVAILABLE, RESERVED, EXPIRED, DEPLETED)

Calculated Fields:

- availableQuantity = quantityOnHand - reservedQuantity

Business Rules:

- Alert when quantity < reorderPoint
- FIFO (First In, First Out) for expiration tracking

j) ConsumableReception

- supplier: String

- purchaseOrderRef: String

- receivingWarehouse: Warehouse

- totalAmount: Decimal

- receivingEmployee: Employee (FK)

- approvalStatus: Enum (PENDING, APPROVED, REJECTED)

- notes: String

- receptionLines: List<ConsReceptionLines>

Relationships:

- Has many ConsReceptionLines

k) ConsReceptionLine

- reception: ConsumableReception (FK)

- consumable: Consumable (FK)

- quantityReceived: Decimal

- unitPrice: Decimal

- batchNumber: String

- qualityStatus: Enum (ACCEPTED, REJECTED, QUARANTINE)

Business Rules:

- Updates ConsumableStock when approved
- Cannot modify after approval

l) TreatmentSheet

- poultryHouse: PoultryHouse (FK)

- lot: ChicksLot (FK)

- veterinarian: Employee (FK)

- diagnosis: String

- treatmentReason: String

- status: Enum (DRAFT, SUBMITED , APPROVED, REJECTED)

- treatmentLines: List<TreatmentSheetLine>

Relationships:

- Has many TreatmentSheetLines

Business Rules:

- One treatment sheet per day per house

- Requires veterinarian approval
- Cannot delete if referenced by consumption

m) TreatmentSheetLine

- treatmentSheet: TreatmentSheet (FK)

- medication: Consumable (FK)

- dosage: Decimal

- dosageUnit: String

- administrationMethod: Enum (WATER, FEED, INJECTION, SPRAY)

- duration: Integer (days)

- startDate: Date

- endDate: Date

- withdrawalPeriod: Integer (days)

- quantityUsed: Decimal

- batchNumber: String

Calculated Fields:

- endDate = startDate + duration
- slaughterAllowedDate = endDate + withdrawalPeriod

Business Rules:

- Auto-creates ConsumptionSheet entries
- Blocks delivery during withdrawal period

n) ConsumptionSheet

- lot: ChicksLot (FK)

- birdsAlive: Integer

- status: Enum (DRAFT, SUBMITED , APPROVED, REJECTED)

- consumptionLines: List<ConsumptionSheetLine>

Relationships:

- Has many ConsumptionSheetLines

Business Rules:

- Auto-created daily for each active lot
- Can edit/delete within 24 hours

- After 24h, requires manager approval to edit

o) ConsumptionSheetLine

- consumptionSheet: ConsumptionSheet (FK)

- consumable: Consumable (FK)

- quantityConsumed: Decimal

- feedingTime: Time

- consumptionPerBird: Decimal (calculated)

- notes: String

- autoGenerated: Boolean

Calculated Fields:

- consumptionPerBird = quantityConsumed / birdsAlive

Business Rules:

- Decrements ConsumableStock
- Validates stock availability

p) ObservationSheet

- weekNumber: Integer

- startDate: Date

- endDate: Date

- lot: ChicksLot (FK)

- observer: Employee (FK)

- status: Enum (DRAFT, SUBMITED, APPROVED, REJECTED)

- startingBirdCount: Integer

- endingBirdCount: Integer

- totalMortality: Integer

- totalFeedConsumed: Integer

- totalWaterConsumed: Integer

- sampleSize: Integer

- averageWeight: Decimal

- weightStdDev: Decimal

- maxWeight: Decimal

- minWeight: Decimal

- weightGain: Decimal

- fcr: Decimal

- adg: Decimal

- healthObs: String

- behavioralNotes: String

- environmentalNotes: String

- concerns: String

Auto-calculated Metrics:

- Starting bird count (from previous week or reception)
- Ending bird count (current - mortality)
- Total mortality count and percentage
- Average daily mortality
- Total feed consumed (sum from ConsumptionSheets)
- Feed Conversion Ratio: totalFeed / totalWeightGain

Manual Input Required:

- Sample weight measurements (random sampling)
- Total water consumed
- Health observations
- Behavioral notes
- Environmental conditions

Business Rules:

- Created weekly (configurable day)
- Previous observation required (except first week)
- Used to populate DeliveryNotice

q) DeliveryNotice

- scheduledDate: DateTime

- farm: BroilerFarm (FK)

- destination: String (Slaughterhouse)

- transportManager: Employee (FK)

- vehicleInfo: String

- approvalStatus: Enum (DRAFT, APPROVED, IN_TRANSIT, DELIVERED)

- approvedBy: Employee (FK)

- dataSource: ENUM(OBSERVATION_SHEET, MANUAL_ENTRY)

- deliveryLines: List<DeliveryNoticeLine>

Relationships:

- Has many DeliveryNoticeLines

r) DeliveryNoticeLine

- deliveryNotice: DeliveryNotice (FK)

- lot: ChicksLot

- estimatedQuantity: Integer

- averageWeight: Decimal (from observation)

- totalEstimatedWeight: Decimal (calculated)

- qualityGrade: Enum (A, B, C)

- specialInstructions: String

- loadingBay: String

Calculated Fields:

- totalEstimatedWeight = estimatedQuantity * averageWeight

Business Rules:

- Cannot create if in withdrawal period
- Requires minimum age (35-42 days typically)
- Requires farm manager approval
- Auto-populated from latest ObservationSheet

s) MortalitySheet

- lot: ChicksLot (FK)

- reportingEmployee: Employee (FK)

- totalMortality: Integer

- cumulativeMortality: Integer

- status: Enum (DRAFT, SUBMITED , APPROVED, REJECTED)

- primaryCause: String

- ageInDays: Integer

- averageWeight: Decimal

- disposalMethod: ENUM(BURIAL, INCINARATION, COMPOSTING)

- locationNotes: String

- observations: String

t) Warehouse

- warehouseName: String

- warehouseCode: String (unique)

- farm: BroilerFarm (FK)

- type: Enum (MAIN_WAREHOUSE, HOUSE_STORAGE, MEDICATION_STORAGE)

- capacity: Decimal

- currentOccupancy: Decimal

- responsibleEmployee: Employee (FK)

- stocks: List<ConsumableStock>

Relationships:

- Belongs to BroilerFarm
- Has many ConsumableStock records
- Has one Employee (worker)

u) User

- username: String (unique)

- passwordHash: String

- employee: Employee (FK, optional)

- lastLogin: DateTime

- failedLoginAttempts: Integer

- accountLocked: Boolean

- mustChangePassword: Boolean

- One-to-One with Employee (optional)
- Separate for system users vs employees